

Predicting locations for bus infrastructure in Philadelphia County

To invest/de-invest into bus infrastructure

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The Problem

What policy problem are you solving?

SEPTA's bus network lacks a data-driven framework to identify which stops and routes need infrastructure investment or should be deprioritized — leading to inefficient resource allocation.

Who is affected and why does it matter?

Philadelphia residents — especially low-income and transit-dependent communities — who rely on buses as their primary mode of transportation. Poor infrastructure directly impacts mobility, equity, and economic access.

What decision would your model inform?

Where SEPTA and city planners should allocate budget for bus infrastructure upgrades (shelters, frequency, stops) and where services may need redesign or reduction.

The Solution

What are you predicting and why?

We are predicting ridership demand across bus stop locations using demographic, spatial, and operational variables — because demand is the strongest indicator of where investment is most needed.

How does prediction help solve the problem?

By quantifying expected ridership at each location, planners can rank stops by need, identify underserved high-demand areas, and justify investment decisions with evidence rather than assumption.

What makes your approach better than current practice?

Current decisions are often based on historical patterns or political priorities. Our model integrates Census demographics, spatial features, and ridership data into a Negative Binomial regression framework — making it reproducible, transparent, and equity-aware.

Data sources

Data Sources Used:

- **Primary Dataset:** SEPTA Bus Ridership Data (2022 to 2025)
- **Secondary Dataset:** Census / ACS Socio-Economic Data /Philadelphia boundary

Why We Are Combining These Sources

✓ Explains transit demand patterns more accurately

We combine bus ridership data with Census/ACS socio-economic data to better explain why ridership varies across locations.

✓ Captures who depends on transit and where

Demographic variables such as population density, income, race, and vehicle ownership help identify transit-dependent communities.

✓ Explains spatial differences in ridership

These variables improve the model by accounting for neighborhood-level variation beyond bus stop locations alone.

✓ Spatially joined using buffer-based linkage

Data is joined spatially using neighborhood or buffer-based linkage around bus stops, representing local catchment areas (e.g., 400–800m radius).

✓ Supports equitable transit planning decisions

This integration makes the model more realistic, policy-relevant, and useful for guiding infrastructure investment decisions.

Methods overview:

We are using **Count Models (Poisson/negative binomial)** for our this model which “Predict appropriate locations for investment or disinvestment in bus infrastructure in Philadelphia County using bus stop and ridership data.” because:

Since bus ridership and related transit activity are **count data**, Poisson or Negative Binomial regression models are appropriate.

- Linear regression is unsuitable because it assumes continuous data.

- Transit ridership data often shows high variation, with some stops being very busy while many others have low usage, leading to uneven distribution. This creates over dispersion (variance greater than the mean), making the Negative Binomial model more suitable than Poisson.

Feature engineering:

We are using the following categorical variables in our model: 1. Weekday vs Weekend
2. High ridership route vs Low ridership route

we are using these variables to:

1. Capture differences in travel behavior
Transit demand changes significantly between weekdays (commuting) and weekends (leisure travel)
2. Represent route-level demand structure
Distinguishes high-demand corridors from low-demand or underperforming routes
3. Improve policy relevance
Helps identify where to invest (high ridership) and where to optimize or redesign service (low ridership)
4. Easy interpretation for planning decisions
Results can be directly translated into operational and infrastructure planning actions

Model Evaluation | Spatial Features

Nearest Transit Stops

Distance to nearest transit stops is used as a spatial predictor, capturing accessibility and connectivity of bus infrastructure across Philadelphia County.

Jobs Within a Half Mile (LEHD LODES)

Employment density within a 0.5-mile buffer is included to capture the influence of job centers on transit demand and ridership patterns.

Roads

Road network data is incorporated to account for physical accessibility and connectivity surrounding each bus stop location.

Model Evaluation | Local Hotspots & Cold Spots

Density of Stops Across Philadelphia

Kernel density estimation is applied to identify spatial clusters of high and low bus stop concentration — revealing hotspots of transit activity and cold spots of underservice across Philadelphia County.

Model Evaluation | Model Validation

Cross Validation

K-fold cross-validation is applied to assess model generalizability by partitioning the data into training and testing sets across multiple folds, ensuring the model performs consistently beyond the training data.

Accuracy Metrics — MAE (Mean Absolute Error)

Mean Absolute Error (MAE) is used to measure the average magnitude of prediction errors, providing an interpretable and policy-relevant measure of model accuracy in predicting ridership counts.

The Evidence | Model Accuracy

How accurate is your model?

- MAE of [X] riders per stop — predictions deviate by an average of [X] passengers
- Model explains approximately [X]% of variation in ridership across Philadelphia County
- Overall predictive performance is [strong/moderate]

The Evidence | Model Validation

How did you validate it?

- Applied **k-fold cross-validation** — dataset split into training and testing sets across [k] folds
- Consistent MAE scores averaging [X] across folds — confirms model generalizes well
- No evidence of **overfitting** to specific locations

Spatial Autocorrelation Diagnostics

- Moran's I on residuals = [X] (p-value = [X])
- Indicates spatial autocorrelation is [negligible/partially addressed]
- Model [adequately captures/does not fully capture] spatial structure of ridership demand

The Evidence | Limitations

What are its limitations?

- Relies on **static Census/ACS data** — may not reflect recent demographic shifts
- Ridership counts are **aggregated at stop level** — do not capture trip purpose or time-of-day variation
- **Weekday/weekend** distinction is the only temporal control included
- Does not account for **real-time service disruptions** or recent SEPTA network changes

The Evidence | Bias Considerations

Could it introduce or perpetuate bias?

- Model uses **race and income** as predictors — risk of reinforcing existing inequities
- Low ridership may reflect **poor service quality** rather than low demand
- Disinvestment decisions could **disproportionately impact** low-income and minority communities
- Model outputs must be interpreted alongside **equity considerations** to ensure fair infrastructure planning

Recommendations

The Implementation | Practical Use

How would the city actually use this?

- Model outputs can be used to generate an **investment priority map** of Philadelphia County bus stops
- Planners can rank stops by **predicted ridership demand** to guide budget allocation
- High-demand locations flagged for **infrastructure upgrades** — shelters, frequency, accessibility improvements
- Low-demand locations flagged for **service redesign** or operational optimization
- Can be integrated into **SEPTA's capital planning cycle** as an evidence-based decision support tool

The Implementation | Data Requirements

What data would they need to collect?

- **Ridership data** — regular updates to SEPTA boarding/alighting counts by stop
- **Census/ACS updates** — annual demographic data at census tract level
- **Land use & employment data** — updated LEHD LODES jobs data within catchment areas
- **Road network data** — OpenStreetMap or PennDOT road updates
- **Infrastructure condition data** — current state of bus stops, shelters, and amenities
- **Real-time service data** — disruptions, route changes, and frequency adjustments

The Implementation | Costs & Benefits

What are the costs?

- **Data collection & maintenance** — ongoing updates to ridership and census data
- **Technical capacity** — GIS and statistical expertise required to run and update the model
- **Integration costs** — embedding model into existing SEPTA planning workflows

What are the benefits?

- **Efficient resource allocation** — directs limited infrastructure budgets to highest-need locations
- **Evidence-based decisions** — reduces reliance on political priorities or ad hoc planning
- **Equity gains** — identifies underserved transit-dependent communities for targeted investment
- **Long-term cost savings** — prevents over-investment in low-demand locations

The Implementation | Safeguards Needed

What safeguards are needed?

- **Equity review** — model outputs must be reviewed against equity metrics before any disinvestment decision
- **Community engagement** — affected neighborhoods should be consulted, especially low-income and minority communities
- **Transparency** — model methodology and results should be publicly accessible and explainable
- **Regular model updates** — predictions should be refreshed with new data to remain accurate and relevant
- **Human oversight** — final investment decisions should involve planners, not be fully automated by the model
- **Bias audits** — periodic checks to ensure model does not systematically disadvantage vulnerable communities

Any Questions & Answers

Thank you for your attention!